Variable Selection for Wasserstein Robust Hypothesis Testing

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Abstract

Variable selection for Wasserstein robust testing admits an exact finite formulation and a maximum-affine dual representation. Dual couplings generate global cuts. The analysis includes error bounds and iteration bounds in terms of solve accuracy and coordinate influence.

1 Problem formulation

Let $[D] = \{1, \dots, D\}$ and let $0 \le d \le D$ be an integer. For a measurable score $f : \mathbb{R}^D \to \mathbb{R}$, define

$$R(f) = \sup_{\substack{P_0 \in \mathcal{U}_0 \\ P_1 \in \mathcal{U}_1}} \left\{ \mathbb{E}_{P_0} \ell(-f(X)) + \mathbb{E}_{P_1} \ell(f(X)) \right\}. \tag{1}$$

The loss ℓ is finite, nonnegative, convex and nondecreasing, with $\ell(0) > 0$. Write $\ell_0(s) = \ell(-s)$ and $\ell_1(s) = \ell(s)$. For $S \subseteq [D]$, let $\mathrm{Proj}_S : \mathbb{R}^D \to \mathbb{R}^{|S|}$ be the coordinate projection and define

$$\mathcal{G}_S = \{ g : \mathbb{R}^{|S|} \to \mathbb{R} : \ g \text{ is Borel measurable} \}.$$

The variable-selection problem is

$$\boxed{V(S) = \inf_{g \in \mathcal{G}_S} R(g \circ \mathrm{Proj}_S), \qquad \mathrm{OPT} = \min_{|S| \le d} V(S).} \tag{2}$$

The N observations are $x_1, \dots, x_N \in \mathbb{R}^D$. Nonempty index sets I_0, I_1 partition $[N]$ by class, with within-class masses $q_i > 0$:

$$\sum_{i \in I_k} q_i = 1, \qquad \widehat{P}_k = \sum_{i \in I_k} q_i \delta_{x_i}, \qquad \mathcal{U}_k = \{ P : W_1(P, \widehat{P}_k) \le \theta_k \}, \quad k = 0, 1.$$

Here δ_x is unit mass at x, $\theta_k > 0$, and W_1 uses ℓ_1 cost on distributions with finite first moments. Unweighted samples have $q_i = 1/|I_k|$ for $i \in I_k$.

2 Dual reformulation

Fix S and write

$$x_i^S = \mathrm{Proj}_S x_i, \qquad \Delta_{ir,j} = |x_{ij} - x_{rj}|, \qquad c_{ir}^S = \|x_i^S - x_r^S\|_1 = \sum_{j \in S} \Delta_{ir,j}.$$

Here $i, r \in [N]$ and $j \in [D]$. A coupling $\pi_{ir} \ge 0$ transports mass from source i to target r. Define

$$\mathcal{T}_k(S) = \left\{ \pi \ge 0 : \sum_r \pi_{ir} = q_i, \quad \sum_{i \in I_k, r} \pi_{ir} c_{ir}^S \le \theta_k \right\}.$$

With target-score variables $u = (u_1, \dots, u_N)$, the finite transport representation is

$$V(S) = \inf_{u \in \mathbb{R}^N} \sum_{k=0}^{1} \max_{\pi \in \mathcal{T}_k(S)} \sum_{i \in I_k, r} \pi_{ir} \ell_k(u_r). \tag{3}$$

Dualizing each transport LP introduces source-loss bounds $a_i \in \mathbb{R}$ and transport prices $\lambda_k \ge 0$. With $J(a, \lambda) = \sum_i q_i a_i + \sum_k \theta_k \lambda_k$, this gives

$$\begin{aligned} V(S) = \min_{a, u \in \mathbb{R}^N, \ \lambda \ge 0} \quad & J(a, \lambda) \\ \text{subject to} \quad & \ell_k(u_r) \le a_i + \lambda_k c_{ir}^S, \quad i \in I_k, \ r \in [N], \ k = 0, 1. \end{aligned} \tag{4}$$

Appendix A proves exactness and recovers an optimal test.

For the binary indicator $z = \mathbf{1}_S$, replace the switched prices $\lambda_k z_j$ by coordinate prices w_{kj}. Since $a_i \geq 0$,

$$\sum_k \theta_k w_k = z \sum_k \theta_k \lambda_k \leq zJ(a,\lambda) \quad \text{when } w_k = \lambda_k z.$$

For fixed $z \in [0,1]^D$, define

$$\begin{aligned} F(z) = \min_{a,u,\lambda,w} \quad & J(a,\lambda) \\ \text{subject to} \quad & \ell_k(u_r) \leq a_i + \sum_j \Delta_{ir,j} w_{kj}, \quad i \in I_k,\ r \in [N],\ k = 0,1, \\ & 0 \leq w_k \leq \lambda_k \mathbf{1}, \qquad k = 0,1, \\ & \theta_0 w_0 + \theta_1 w_1 \leq zJ(a,\lambda), \\ & \lambda \geq 0, \qquad a,u \in \mathbb{R}^N. \end{aligned} \tag{5}$$

Here $w_k = (w_{k1}, \ldots, w_{kD})^\top$ and $\mathbf{1} = (1, \ldots, 1)^\top$; vector inequalities are coordinatewise. For binary z, under the other constraints,

$$\theta_0 w_0 + \theta_1 w_1 \leq zJ \quad \Leftrightarrow \quad w_{kj} = 0 \quad \text{for } z_j = 0,\ k = 0,1.$$

When $z_j = 1$, that particular inequality is redundant and may be omitted:

$$\theta_0 w_{0j} + \theta_1 w_{1j} \leq \sum_k \theta_k \lambda_k \leq J.$$

Deleting the entire family also removes the zero-coordinate restrictions, giving $V([D])$ for every z. Retaining it gives $F(\mathbf{1}_S) = V(S)$; Proposition B.1(i) proves equality of the projected feasible sets and preservation of the objective.

A dual of (5), with a scaled transport coupling $\Pi \geq 0$, a scalar $\alpha \geq 0$, and coordinate multipliers $\mu \geq 0$, is

$$\begin{aligned} F(z) = \max_{\Pi,\alpha,\mu} \quad & H(\Pi) \\ \text{subject to} \quad & (\Pi,\alpha,\mu) \in \mathcal{D}, \qquad \alpha + \mu^\top z = 1. \end{aligned} \tag{6}$$

Here

$$H(\Pi) = \sum_{r=1}^{N} \inf_{s \in \mathbb{R}} \sum_{k=0}^{1} \sum_{i \in I_k} \Pi_{ir} \ell_k(s),$$

and, with $(v)_+ = \max\{v, 0\}$,

$$\mathcal{D} = \left\{ (\Pi,\alpha,\mu) \geq 0 : \begin{array}{l} \displaystyle\sum_r \Pi_{ir} = \alpha q_i, \quad i \in [N], \\ \displaystyle\sum_j \left(\sum_{i \in I_k, r} \Pi_{ir} \Delta_{ir,j} - \theta_k \mu_j \right)_+ \leq \alpha \theta_k, \quad k = 0,1 \end{array} \right\}. \tag{7}$$

The coupling has source masses αq_i; H minimizes its weighted loss over one score per target. The Lagrangian derivation is in Proposition B.1(i).

For a normalized tuple $\alpha + \mu^\top z = 1$, scaling all three dual variables by $\tau \geq 0$ preserves $\mathcal{D}$ and gives

$$H(\tau\Pi) = \tau H(\Pi), \qquad \max_{\tau \geq 0} \{2\sqrt{\tau H(\Pi)} - \tau\} = H(\Pi), \quad \tau^* = H(\Pi).$$

Optimizing this scale on every ray removes the normalization:

$$\boxed{F(z) = \max_{(\Pi,\alpha,\mu) \in \mathcal{D}} \{2\sqrt{H(\Pi)} - \alpha - \mu^\top z\}.} \tag{8}$$

Proposition B.1(ii) proves this equivalence and attainment. A normalized optimizer $(\bar{\Pi}, \bar{\alpha}, \bar{\mu})$ gives the affine optimizer $F(z)(\bar{\Pi}, \bar{\alpha}, \bar{\mu})$; the scaling must also be applied to the cut slope.

3 Cutting-plane algorithm

Let $\mathcal{Z}_d = \{z \in \{0,1\}^D : \mathbf{1}^\top z \le d\}$. At z^t, evaluate $F(z^t)$ by (8) and take an optimizer (Π^t, α^t, μ^t). It gives a global affine lower bound attained at z^t:

$$\begin{aligned} &c_t(z) := 2\sqrt{H(\Pi^t)} - \alpha^t - (\mu^t)^\top z, \\ &F(z) \ge c_t(z) = F(z^t) - (\mu^t)^\top (z - z^t), \qquad c_t(z^t) = F(z^t). \end{aligned} \tag{9}$$

The next selection and lower bound solve

$$\begin{aligned} (z^{t+1}, \eta^{t+1}) \in \underset{z \in \mathcal{Z}_d,\ \eta \ge 0}{\arg\min} \quad & \eta \\ \text{subject to} \quad & \eta \ge F(z^s) - (\mu^s)^\top (z - z^s), \qquad s = 1, \dots, t. \end{aligned} \tag{10}$$

For the iteration bound, define

$$C = 2\ell(0), \qquad A_{kj} = \sum_{i \in I_k} q_i \max_{r \in [N]} \Delta_{ir,j}, \qquad M_j = C \max_{k=0,1} \frac{A_{kj}}{\theta_k}. \tag{11}$$

Order these bounds as $M_{(1)} \ge \cdots \ge M_{(D)}$ and put, for $m = 0, \dots, D$,

$$\rho_m = \sum_{r=m+1}^{\min(D, m+d)} M_{(r)}, \qquad K_m = \sum_{r=0}^{\min(d,m)} \binom{m}{r}. \tag{12}$$

Here ρ_m bounds the total influence of at most d coordinates outside the first m, and K_m counts the possible selections on those first m coordinates.

For inexact solves, retain all global affine lower bounds $c_s(z) \le F(z)$ and let p_s be their paired primal feasible values. Define

$$\begin{aligned} U_t = \min_{s \le t} p_s, \qquad & m_t(z) = \max\{0, c_1(z), \dots, c_t(z)\}, \\ \underline{L}_t \le \min_{z \in \mathcal{Z}_d} m_t(z), \qquad & L_t = \max\{0, \underline{L}_1, \dots, \underline{L}_t\}, \qquad G_t = U_t - L_t. \end{aligned}$$

Thus $\underline{L}_t$ is a global master lower bound and G_t is the computed optimality gap. Let f_t be the recovered score paired with U_t.

Proposition 3.1 (Convergence)**.** (i) *At every $z \in [0,1]^D$, adding $\alpha \le C$ and $0 \le \mu_j \le M_j$ to the affine dual preserves its attained optimal value $F(z)$.*

(ii) *Fix $\delta, \beta \ge 0$. Suppose the returned affine lower bounds have slopes $-\mu^s$, and each inner solve and master solve satisfies*

$$\begin{aligned} 0 \le \mu^s \le M, \qquad & 0 \le p_s - c_s(z^s) \le \delta, \\ z^{t+1} \in \mathcal{Z}_d, \qquad & \eta^{t+1} \ge m_t(z^{t+1}), \qquad 0 \le \eta^{t+1} - L_t \le \beta. \end{aligned}$$

For every completed iteration t and every m with $t \ge K_m$,

$$0 \le R(f_t) - \mathrm{OPT} \le G_t \le \delta + \beta + \rho_m. \tag{13}$$

For $\varepsilon \ge \delta + \beta$ with $\varepsilon > 0$, stopping at $G_t \le \varepsilon$ requires at most

$$T_\varepsilon \le K_{m_\varepsilon}, \qquad m_\varepsilon = \min\{m \in \{0, \dots, D\} : \rho_m \le \varepsilon - \delta - \beta\}. \tag{14}$$

Exact solves correspond to $\delta = \beta = 0$ and $L_t = \eta^{t+1}$. The proof is in Appendix C.

Algorithm 1 Exact cutting-plane method for variable selection

Require: Problem data, budget d, tolerance $\varepsilon > 0$, and $z^1 \in \mathcal{Z}_d$.
Ensure: A subset S^*, its optimal test f^*, and a risk interval of width at most ε.

$t \leftarrow 1, U \leftarrow +\infty$.
while true **do**
  Solve (8) exactly at z^t; choose μ^t with $0 \le \mu_j^t \le M_j$ as in Proposition 3.1(i).
  if $F(z^t) < U$ **then**
    $(U, z^*) \leftarrow (F(z^t), z^t)$.
  end if
  Retain the cut (9) with index t.
  Solve (10) exactly to obtain (z^{t+1}, η^{t+1}).
  if $U - \eta^{t+1} \le \varepsilon$ **then**
    $S^* \leftarrow \{j : z_j^* = 1\}$.
    Solve (4) for S^*; construct g^* by (20) and set $f^* = g^* \circ \mathrm{Proj}_{S^*}$.
    return $(S^*, f^*, [\eta^{t+1}, U])$.
  end if
  $t \leftarrow t + 1$.
end while

A Proof of the finite representation

Let $x_i \in \mathbb{R}^D$, $i \in [N]$, and let nonempty sets I_0, I_1 partition $[N]$. Let $q_i > 0$, $\sum_{i \in I_k} q_i = 1$, and $\theta_k > 0$. The loss $\ell : \mathbb{R} \to [0, \infty)$ is finite, convex and nondecreasing, with $\ell(0) > 0$; put $\ell_0(s) = \ell(-s)$ and $\ell_1(s) = \ell(s)$. For $S \subseteq [D]$, let Proj_S be coordinate projection; δ_x denotes unit mass at x. Define

$$x_i^S = \mathrm{Proj}_S\, x_i, \qquad c_{ir}^S = \|x_i^S - x_r^S\|_1, \qquad \widehat{P}_k = \sum_{i \in I_k} q_i \delta_{x_i},$$

$$\mathcal{U}_k = \{P : W_1(P, \widehat{P}_k) \le \theta_k\}, \qquad \mathcal{G}_S = \{g : \mathbb{R}^{|S|} \to \mathbb{R} : \ g \text{ is Borel measurable}\},$$

$$R(f) = \sum_{k=0}^{1} \sup_{P \in \mathcal{U}_k} \mathbb{E}_P \ell_k(f(X)), \qquad V(S) = \inf_{g \in \mathcal{G}_S} R(g \circ \mathrm{Proj}_S).$$

Here W_1 uses ℓ_1 transport cost and finite-first-moment distributions. Set

$$T(S) = \inf_{a, u \in \mathbb{R}^N,\ \lambda \in \mathbb{R}_+^2} \quad J(a, \lambda) := \sum_i q_i a_i + \sum_{k=0}^{1} \theta_k \lambda_k \qquad (15)$$
$$\text{subject to} \quad \ell_k(u_r) \le a_i + \lambda_k c_{ir}^S, \quad i \in I_k,\ r \in [N],\ k = 0, 1.$$

$T(S)$ is the finite value; equality with $V(S)$ is proved below.

Proposition A.1. *For every $S \subseteq [D]$, $V(S) = T(S)$, the infimum defining $T(S)$ is attained, and each optimal finite solution produces a Borel score $g^* : \mathbb{R}^{|S|} \to \mathbb{R}$ with $R(g^* \circ \mathrm{Proj}_S) = V(S)$.*

Proof. Finite transport duality. Fix $u \in \mathbb{R}^N$. Define

$$\mathcal{T}_k(S) = \left\{ \pi \in \mathbb{R}_+^{|I_k| \times N} : \sum_r \pi_{ir} = q_i \ (i \in I_k), \quad \sum_{i \in I_k, r} \pi_{ir} c_{ir}^S \le \theta_k \right\}.$$

With $a_i \in \mathbb{R}$, $\lambda_k \geq 0$, and indices $i \in I_k$, $r \in [N]$,

$$\begin{aligned} L_k(\pi, a, \lambda_k; u) &= \sum_{i,r} \pi_{ir} \ell_k(u_r) + \sum_i a_i \left(q_i - \sum_r \pi_{ir} \right) + \lambda_k \left(\theta_k - \sum_{i,r} \pi_{ir} c_{ir}^S \right) \\ &= \sum_i q_i a_i + \theta_k \lambda_k + \sum_{i,r} \pi_{ir} [\ell_k(u_r) - a_i - \lambda_k c_{ir}^S]. \end{aligned} \tag{16}$$

$$\sup_{\pi \geq 0} L_k = \begin{cases} \sum_i q_i a_i + \theta_k \lambda_k, & \ell_k(u_r) \leq a_i + \lambda_k c_{ir}^S \quad \forall i, r, \\ +\infty, & \text{otherwise.} \end{cases}$$

LP strong duality gives

$$\begin{aligned} \max_{\pi \in \mathcal{T}_k(S)} \sum_{i \in I_k, r} \pi_{ir} \ell_k(u_r) = \min_{(a_i)_{i \in I_k},\, \lambda_k \geq 0} \quad & \sum_{i \in I_k} q_i a_i + \theta_k \lambda_k \\ \text{subject to} \quad & \ell_k(u_r) \leq a_i + \lambda_k c_{ir}^S, \quad i \in I_k,\ r \in [N]. \end{aligned} \tag{17}$$

$$T(S) = \inf_{u \in \mathbb{R}^N} \sum_{k=0}^{1} \max_{\pi \in \mathcal{T}_k(S)} \sum_{i \in I_k, r} \pi_{ir} \ell_k(u_r).$$

Only the finite LP has been dualized; no minimax exchange is used.

Lower bound: $T(S) \leq V(S)$. For $g \in \mathcal{G}_S$ and $\pi \in \mathcal{T}_k(S)$, choose $\tilde{x}_{ir} \in \mathbb{R}^D$ with

$$\begin{gathered} \mathrm{Proj}_S \tilde{x}_{ir} = x_r^S, \qquad \mathrm{Proj}_{S^c} \tilde{x}_{ir} = \mathrm{Proj}_{S^c} x_i, \qquad \|\tilde{x}_{ir} - x_i\|_1 = c_{ir}^S. \\ P_k^\pi = \sum_{i \in I_k, r} \pi_{ir} \delta_{\tilde{x}_{ir}}, \qquad W_1(P_k^\pi, \widehat{P}_k) \leq \sum_{i \in I_k, r} \pi_{ir} c_{ir}^S \leq \theta_k, \\ \mathbb{E}_{P_k^\pi} \ell_k(g(\mathrm{Proj}_S X)) = \sum_{i \in I_k, r} \pi_{ir} \ell_k(g(x_r^S)). \end{gathered} \tag{18}$$

$$\begin{aligned} V(S) &\geq \inf_{g \in \mathcal{G}_S} \sum_k \max_{\pi \in \mathcal{T}_k(S)} \sum_{i \in I_k, r} \pi_{ir} \ell_k(g(x_r^S)) \\ &\geq \inf_{u \in \mathbb{R}^N} \sum_k \max_{\pi \in \mathcal{T}_k(S)} \sum_{i \in I_k, r} \pi_{ir} \ell_k(u_r) = T(S). \end{aligned} \tag{19}$$

Upper bound: $V(S) \leq T(S)$. For any finite feasible (a, u, λ) and $y \in \mathbb{R}^{|S|}$, set

$$\begin{gathered} E_k(y) = \min_{i \in I_k} \{a_i + \lambda_k \|x_i^S - y\|_1\}, \qquad i_k(y) = \min \arg\min_{i \in I_k} \{a_i + \lambda_k \|x_i^S - y\|_1\}, \\ r_k = \|x_{i_k}^S - y\|_1, \qquad \rho = \|x_{i_0}^S - x_{i_1}^S\|_1 \leq r_0 + r_1. \\ g(y) = \begin{cases} (1-\gamma) u_{i_0} + \gamma u_{i_1}, & \gamma = \min\{1, r_0/\rho\}, \quad \rho > 0, \\ u_{i_0}, & \rho = 0. \end{cases} \end{gathered} \tag{20}$$

For $\rho > 0$, $\gamma\rho \leq r_0$ and $(1-\gamma)\rho \leq r_1$. Convexity and finite feasibility yield

$$\begin{aligned} \ell_0(g(y)) &\leq (1-\gamma)\ell_0(u_{i_0}) + \gamma \ell_0(u_{i_1}) \leq a_{i_0} + \lambda_0 \gamma \rho \leq E_0(y), \\ \ell_1(g(y)) &\leq (1-\gamma)\ell_1(u_{i_0}) + \gamma \ell_1(u_{i_1}) \leq a_{i_1} + \lambda_1 (1-\gamma)\rho \leq E_1(y). \end{aligned}$$

For $\rho = 0$, the source–target pairs (i_0, i_0) and (i_1, i_0) give the same two bounds. The finite tie rule makes g Borel, with $|g(y)| \leq \max_r |u_r|$; interpolation of the u_r is unnecessary. For $P \in \mathcal{U}_k$, disintegrate an optimal coupling as source-indexed measures ν_i (splitting repeated sources by q_i):

$$\nu_i(\mathbb{R}^D) = q_i, \qquad \sum_{i \in I_k} \nu_i = P, \qquad \sum_{i \in I_k} \int \|x - x_i\|_1 \, \nu_i(dx) \leq \theta_k.$$

$$\begin{aligned}\mathbb{E}_P \ell_k(g(\mathrm{Proj}_S X)) &\le \sum_{i\in I_k} \int [a_i + \lambda_k \| \mathrm{Proj}_S\, x - x_i^S\|_1]\, \nu_i(dx) \\ &\le \sum_{i\in I_k} q_i a_i + \theta_k \lambda_k.\end{aligned} \tag{21}$$

$$R(g \circ \mathrm{Proj}_S) \le J(a,\lambda), \qquad V(S) \le T(S).$$

The feasible point $(a,u,\lambda) = (\ell(0)\mathbf{1}, 0, 0)$ has value $C = 2\ell(0)$. On the closed sublevel $J \le C$,

$$0 \le a_i \le C/q_i, \qquad 0 \le \lambda_k \le C/\theta_k.$$

The loss constraints bound both $\ell(u_r)$ and $\ell(-u_r)$. If ℓ is nonconstant, $\ell(t) \to \infty$ as $t \to \infty$, so u is bounded; if ℓ is constant, restrict to $u = 0$. Weierstrass's theorem gives a minimizer, whose extension attains $V(S)$. □

B Proof of the dual reformulation

Let $x_i \in \mathbb{R}^D$, let nonempty I_0, I_1 partition $[N]$, and let $q_i, \theta_k > 0$ with $\sum_{i\in I_k} q_i = 1$. Let $\ell : \mathbb{R} \to [0,\infty)$ be finite, convex and nondecreasing, with $\ell(0) > 0$; write $\ell_0(s) = \ell(-s)$ and $\ell_1(s) = \ell(s)$. Put

$$\Delta_{ir,j} = |x_{ij} - x_{rj}|, \qquad c_{ir}^S = \sum_{j\in S} \Delta_{ir,j}, \qquad J(a,\lambda) = \sum_i q_i a_i + \sum_{k=0}^{1} \theta_k \lambda_k.$$

For $S \subseteq [D]$ and $z \in [0,1]^D$, consider

$$\begin{aligned} V(S) = \min_{a,u,\lambda} \quad & J(a,\lambda) \\ \text{subject to} \quad & \ell_k(u_r) \le a_i + \lambda_k c_{ir}^S, \quad i \in I_k,\ r \in [N],\ k = 0,1, \\ & a, u \in \mathbb{R}^N, \qquad \lambda \in \mathbb{R}_+^2, \end{aligned} \tag{22}$$

$$\begin{aligned} F(z) = \min_{a,u,\lambda,w} \quad & J(a,\lambda) \\ \text{subject to} \quad & \ell_k(u_r) \le a_i + \sum_{j=1}^{D} \Delta_{ir,j} w_{kj}, \quad i \in I_k,\ r \in [N],\ k = 0,1, \\ & 0 \le w_{kj} \le \lambda_k, \quad k = 0,1,\ j \in [D], \\ & \sum_{k=0}^{1} \theta_k w_{kj} \le z_j J(a,\lambda), \quad j \in [D], \\ & a, u \in \mathbb{R}^N, \qquad \lambda \in \mathbb{R}_+^2. \end{aligned} \tag{23}$$

For $\Pi \in \mathbb{R}_+^{N\times N}$, define

$$H(\Pi) = \sum_{r=1}^{N} \inf_{s\in\mathbb{R}} \sum_{k=0}^{1} \sum_{i\in I_k} \Pi_{ir} \ell_k(s), \qquad d_{kj}(\Pi) = \sum_{i\in I_k, r} \Pi_{ir} \Delta_{ir,j}.$$

$$\mathcal{D} = \left\{ (\Pi,\alpha,\mu) \ge 0 : \begin{array}{l} \sum_r \Pi_{ir} = \alpha q_i \quad (i \in [N]), \\ \sum_j (d_{kj}(\Pi) - \theta_k \mu_j)_+ \le \alpha\theta_k \quad (k = 0,1) \end{array} \right\}, \qquad (v)_+ = \max\{v, 0\}. \tag{24}$$

Proposition B.1 (Dual representations and subgradients)**.** (i) *For every $S \subseteq [D]$, $F(\mathbf{1}_S) = V(S)$. For every $z \in [0,1]^D$, the primal minimum and the normalized dual maximum are attained, and*

$$0 < F(z) = \max_{\substack{(\Pi,\alpha,\mu)\in\mathcal{D} \\ \alpha + \mu^\top z = 1}} H(\Pi) \le 2\ell(0). \tag{25}$$

(ii) *The same value is attained by the affine representation*

$$F(z) = \max_{(\Pi,\alpha,\mu)\in\mathcal{D}} \{2\sqrt{H(\Pi)} - \alpha - \mu^\top z\}. \tag{26}$$

For any normalized optimizer $(\bar{\Pi}, \bar{\alpha}, \bar{\mu})$ *at* z*, an affine optimizer is*

$$(\Pi, \alpha, \mu) = F(z)(\bar{\Pi}, \bar{\alpha}, \bar{\mu}), \qquad H(\Pi) = F(z)^2. \tag{27}$$

(iii) *The function* F *is convex. For any affine optimizer* (Π^t, α_t, μ^t) *at* z^t*, its affine function satisfies*

$$\begin{aligned} c_t(z) &:= 2\sqrt{H(\Pi^t)} - \alpha_t - (\mu^t)^\top z \le F(z) \quad (z \in [0,1]^D), \\ c_t(z^t) &= F(z^t), \qquad c_t(z) = F(z^t) - (\mu^t)^\top (z - z^t). \end{aligned} \tag{28}$$

Thus $-\mu^t \in \partial F(z^t)$*, with subgradients understood on* $[0,1]^D$*; differentiability is not assumed.*

Proof. (i) Diagonal feasibility gives $a_i \ge \ell_k(u_i) \ge 0$. For a feasible triple in (22), let $z = \mathbf{1}_S$ and $w_k = \lambda_k z$. Then

$$\ell_k(u_r) \le a_i + \sum_j \Delta_{ir,j} w_{kj}, \qquad \sum_k \theta_k w_{kj} = z_j \sum_k \theta_k \lambda_k \le z_j J(a,\lambda).$$

Conversely, (23) at $z = \mathbf{1}_S$ gives

$$\begin{aligned} w_{kj} = 0 \quad (j \notin S), \qquad 0 \le w_{kj} \le \lambda_k \quad (j \in S), \\ \ell_k(u_r) \le a_i + \sum_{j\in S} \Delta_{ir,j} w_{kj} \le a_i + \lambda_k c^S_{ir}. \end{aligned}$$

Both maps preserve J, proving binary equivalence.

Assign multipliers $\Pi_{ir}, \beta_{kj}, \mu_j \ge 0$ to the loss, price-cap and selector inequalities of (23). Keep a, u free and $\lambda, w \ge 0$; put $\alpha = 1 - \mu^\top z$. The Lagrangian is

$$\begin{aligned} \mathcal{L} &= J + \sum_{k,i\in I_k,r} \Pi_{ir} \left[\ell_k(u_r) - a_i - \sum_j \Delta_{ir,j} w_{kj} \right] + \sum_{k,j} \beta_{kj}(w_{kj} - \lambda_k) \\ &\quad + \sum_j \mu_j \left[\sum_k \theta_k w_{kj} - z_j J \right] \\ &= \sum_i a_i \left(\alpha q_i - \sum_r \Pi_{ir} \right) + \sum_{r,k,i\in I_k} \Pi_{ir} \ell_k(u_r) \\ &\quad + \sum_k \lambda_k \left(\alpha\theta_k - \sum_j \beta_{kj} \right) + \sum_{k,j} w_{kj} [\beta_{kj} + \theta_k \mu_j - d_{kj}(\Pi)]. \end{aligned} \tag{29}$$

Its infimum over a, λ, w is finite precisely when

$$\sum_r \Pi_{ir} = \alpha q_i, \qquad \sum_j \beta_{kj} \le \alpha\theta_k, \qquad \beta_{kj} \ge (d_{kj}(\Pi) - \theta_k \mu_j)_+. \tag{30}$$

The row sums force $\alpha \ge 0$. By separability over target columns,

$$\begin{aligned} \inf_{u\in\mathbb{R}^N} \sum_{r=1}^N \sum_{k=0}^1 \sum_{i\in I_k} \Pi_{ir} \ell_k(u_r) &= \sum_{r=1}^N \inf_{s\in\mathbb{R}} \sum_{k=0}^1 \sum_{i\in I_k} \Pi_{ir} \ell_k(s) \\ &= H(\Pi). \end{aligned} \tag{31}$$

Eliminating β is exact because its smallest feasible entries are $\beta_{kj} = (d_{kj}(\Pi) - \theta_k \mu_j)_+$. This gives $\mathcal{D}$ and $\alpha + \mu^\top z = 1$.

For $J_0 = \{j : z_j = 0\}$, feasibility forces $w_{kj} = 0$. After deleting these variables and selector rows, a strict feasible point is

$$u = 0, \qquad a_i = \ell(0) + 1, \qquad \lambda_k = 1, \qquad w_{kj} = z_j/2 \quad (j \notin J_0).$$

The loss and price-cap inequalities are strict, and

$$\sum_k \theta_k w_{kj} = \frac{z_j}{2} \sum_k \theta_k < z_j \left[2(\ell(0) + 1) + \sum_k \theta_k \right] = z_j J \quad (j \notin J_0).$$

Slater's theorem gives strong duality and dual attainment for the reduced problem, also when $J_0 = [D]$. Restore the deleted coordinates with

$$\beta_{kj} = 0, \qquad \mu_j = \max_{k=0,1} d_{kj}(\Pi)/\theta_k \quad (j \in J_0).$$

These choices satisfy (30) and change neither the objective nor normalization.

The constant feasible point $(a, u, \lambda, w) = (\ell(0)\mathbf{1}, 0, 0, 0)$ has value $C = 2\ell(0)$. Its closed sublevel $J \le C$ satisfies

$$0 \le a_i \le C/q_i, \qquad 0 \le w_{kj} \le \lambda_k \le C/\theta_k,$$
$$\ell_k(u_r) \le C/q_i + (C/\theta_k) \sum_j \Delta_{ir,j} \quad (i \in I_k,\ r \in [N],\ k = 0, 1).$$

For nonconstant ℓ, convexity and monotonicity give $\ell(t) \to \infty$ as $t \to \infty$, so the two losses bound u on both sides. For constant ℓ, take $u = 0$. A compact restriction therefore contains a primal minimizer. If its value were zero, $a = \lambda = w = 0$ would imply

$$0 < 2\ell(0) \le \ell(-u_r) + \ell(u_r) \le 0,$$

a contradiction. This proves (25) and primal attainment; binary equivalence gives attainment of (22).

(ii) The set $\mathcal{D}$ is a cone and $H(\tau\Pi) = \tau H(\Pi) \ge 0$ for $\tau \ge 0$. For a feasible tuple put $b = \alpha + \mu^\top z$. If $b > 0$, normalization and (i) give $H(\Pi)/b \le F(z)$, hence

$$2\sqrt{H(\Pi)} - b = \frac{H(\Pi)}{b} - \left(\sqrt{\frac{H(\Pi)}{b}} - \sqrt{b} \right)^2 \le F(z). \tag{32}$$

If $b = 0$, then $\alpha = 0, \Pi = 0$, and the affine value is zero. Conversely, any normalized optimizer satisfies

$$H(F(z)\bar{\Pi}) = F(z)^2, \qquad F(z)(\bar{\alpha} + \bar{\mu}^\top z) = F(z), \qquad 2\sqrt{H(F(z)\bar{\Pi})} - F(z) = F(z).$$

Thus (26) is attained by (27).

(iii) The feasible set $\mathcal{D}$ does not depend on z; (26) is a maximum of affine functions and hence convex. Dual feasibility gives $c_t(z) \le F(z)$, and optimality at z^t gives $c_t(z^t) = F(z^t)$. Subtracting these values yields (28) and the stated subgradient inequality. The normalized multiplier must be scaled as $\mu^t = F(z^t)\bar{\mu}^t$.

For an inexact solve, let $(\Pi, \alpha, \mu) \in \mathcal{D}$, $0 \le h \le H(\Pi)$ and $c(z) = 2\sqrt{h} - \alpha - \mu^\top z$. If p_t is primal feasible at z^t and $p_t - c(z^t) \le \delta$, then

$$c(z) \le F(z), \qquad 0 \le F(z^t) - c(z^t) \le p_t - c(z^t) \le \delta,$$
$$F(z) \ge F(z^t) - \mu^\top (z - z^t) - \delta \quad (z \in [0, 1]^D).$$

□

C Proof of convergence

Let $x_i \in \mathbb{R}^D$, let nonempty I_0, I_1 partition $[N]$, and let $q_i, \theta_k > 0$ with $\sum_{i \in I_k} q_i = 1$. The loss $\ell : \mathbb{R} \to [0, \infty)$ is finite, convex and nondecreasing, with $\ell(0) > 0$. Let $d \in \{0, \dots, D\}$. Write

$$F(z) = \max_{(\Pi,\alpha,\mu)\in\mathcal{D}} \{2\sqrt{H(\Pi)} - \alpha - \mu^\top z\}, \qquad \mathcal{Z}_d = \{z \in \{0,1\}^D : \mathbf{1}^\top z \le d\}, \qquad \mathrm{OPT} = \min_{z\in\mathcal{Z}_d} F(z),$$

where H and $\mathcal{D}$ are defined in Appendix B. With $\Delta_{ir,j} = |x_{ij} - x_{rj}|$, set

$$C = 2\ell(0), \qquad A_{kj} = \sum_{i\in I_k} q_i \max_{r\in[N]} \Delta_{ir,j}, \qquad M_j = C \max_{k=0,1} \frac{A_{kj}}{\theta_k}.$$

Let J_m be the indices of the m largest M_j, breaking ties by index. For $M_{(1)} \ge \cdots \ge M_{(D)}$, define

$$\rho_m = \sum_{r=m+1}^{\min(D,m+d)} M_{(r)}, \qquad K_m = \sum_{r=0}^{\min(d,m)} \binom{m}{r}, \qquad 0 \le m \le D.$$

Fix $\delta, \beta \ge 0$ and $\varepsilon > 0$. At each evaluated $z^s = \mathbf{1}_{S_s}$, take a feasible primal value p_s and recover $f_s = g_s \circ \mathrm{Proj}_{S_s}$, so that $R(f_s) \le p_s$. The paired affine lower bound is

$$c_s(z) = 2\sqrt{h_s} - \alpha_s - (\mu^s)^\top z, \qquad (\Pi^s, \alpha_s, \mu^s) \in \mathcal{D}, \qquad 0 \le h_s \le H(\Pi^s),$$
$$0 \le \mu^s \le M, \qquad 0 \le p_s - c_s(z^s) \le \delta.$$

Retain all cuts and define

$$m_t(z) = \max\{0, c_1(z), \dots, c_t(z)\}, \qquad U_t = \min_{s\le t} p_s, \qquad f_t = f_{s_t}, \quad s_t \in \arg\min_{s\le t} p_s,$$
$$\underline{L}_t \le \min_{z\in\mathcal{Z}_d} m_t(z), \qquad L_t = \max\{0, \underline{L}_1, \dots, \underline{L}_t\}, \qquad G_t = U_t - L_t.$$

The master returns $z^{t+1} \in \mathcal{Z}_d$ and η^{t+1} satisfying

$$\eta^{t+1} \ge m_t(z^{t+1}), \qquad 0 \le \eta^{t+1} - L_t \le \beta.$$

Proof of Proposition 3.1. (i) Scale a normalized optimizer by $F(z)$ as in Appendix B. Since $\bar\alpha + \bar\mu^\top z = 1$ and $F(z) \le C$,

$$0 \le \alpha = F(z)\bar\alpha \le C, \qquad d_{kj}(\Pi) = \sum_{i\in I_k, r} \Pi_{ir}\Delta_{ir,j} \le \alpha \sum_{i\in I_k} q_i \max_r \Delta_{ir,j} = \alpha A_{kj}.$$

Put $b_j = \max_k d_{kj}(\Pi)/\theta_k$ and $\mu'_j = \min\{\mu_j, b_j\}$. Whenever $\mu_j > b_j$,

$$(d_{kj}(\Pi) - \theta_k\mu'_j)_+ = (d_{kj}(\Pi) - \theta_k\mu_j)_+ = 0 \quad (k = 0,1).$$

Thus $(\Pi, \alpha, \mu') \in \mathcal{D}$ and

$$F(z) \ge 2\sqrt{H(\Pi)} - \alpha - (\mu')^\top z \ge 2\sqrt{H(\Pi)} - \alpha - \mu^\top z = F(z),$$
$$0 \le \mu'_j \le b_j \le \alpha \max_k A_{kj}/\theta_k \le M_j.$$

The bounded dual therefore retains an optimizer and the same value.
(ii) Primal feasibility, dual feasibility and $h_s \le H(\Pi^s)$ give

$$0 \le m_t \le F, \qquad L_t \le \min_{z\in\mathcal{Z}_d} m_t(z) \le \mathrm{OPT} \le R(f_t) \le U_t. \tag{33}$$

Because all cuts are retained, m_t and L_t are nondecreasing, whereas U_t and G_t are nonincreasing. For $s < r \le t+1$, the master point $z^r = \mathbf{1}_{S_r}$ satisfies

$$\begin{aligned} G_{r-1} &\le p_s - c_s(z^r) + \eta^r - L_{r-1} \\ &= p_s - c_s(z^s) + (\mu^s)^\top (z^r - z^s) + \eta^r - L_{r-1} \\ &\le \delta + \beta + \sum_{j \in S_r \setminus S_s} M_j. \end{aligned} \tag{34}$$

Fix m and suppose $G_t > \delta + \beta + \rho_m$. Monotonicity and (34) imply

$$\sum_{j \in S_r \setminus S_s} M_j > \rho_m \qquad (s < r \le t+1).$$

If $S_r \cap J_m = S_s \cap J_m$, then $S_r \setminus S_s \subseteq [D] \setminus J_m$ and $|S_r \setminus S_s| \le d$, so

$$\sum_{j \in S_r \setminus S_s} M_j \le \sum_{q=m+1}^{\min(D, m+d)} M_{(q)} = \rho_m,$$

a contradiction. Thus $S_1 \cap J_m, \dots, S_{t+1} \cap J_m$ are pairwise distinct, but there are only K_m such subsets. Hence $t+1 \le K_m$. For $t \ge K_m$ this is impossible, proving

$$0 \le R(f_t) - \mathrm{OPT} \le G_t \le \delta + \beta + \rho_m \qquad (t \ge K_m).$$

For $\varepsilon \ge \delta + \beta$, set $m_\varepsilon = \min\{m \in \{0, \dots, D\} : \rho_m \le \varepsilon - \delta - \beta\}$. This gives $G_{K_{m_\varepsilon}} \le \varepsilon$ and $T_\varepsilon \le K_{m_\varepsilon}$, unless the method stopped earlier. The last master candidate is included in the counting argument without an additional inner solve. In particular,

$$\rho_D = 0, \quad K_D = \sum_{r=0}^{d} \binom{D}{r}, \qquad d = 0 \text{ or } M = 0 \quad \Longrightarrow \quad G_1 \le \delta + \beta.$$

□